Verification Continuum™

HAPS®-200 12F

Reference Manual

HW1333-0
July 2026 V1



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Revision History

Date	Revision	Comment
Mar 2026	1	Initial version
Jul 2026	2	Page 26 - Added available DDR4 Memory. Page 28 - Added 1.1V to supported VCCO voltages.

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Free and Open-Source Licensing Notices 1-2

Revision History 2-5

Chapter 1: General Information

Emissions 1-9
Support Contact Information 1-10
IMPORTANT! 1-11

Overview 1-14
Features 1-14
High level Construction 1-15
FPGA modules and AMD Versal Premium Technology 1-15
Layout 1-16
Block Diagram 1-17

I/O Signals and Connections 1-18
LEDs 1-19
Clocks 1-20

Chapter 2: Concept

HapsTrak 2-21

FPGA Inter-connects 2-23
GTYP 2-23
XPIO 2-24
Interconnect Boards and Cables 2-24

Extension Boards 2-25

Multi-Module Setups 2-25

Available DDR4 Memory 2-26

Chapter 3: Expansion and I/Os

I/O Connectors 3-27
HapsTrak 4 3-27
HapsTrak 4 Signals 3-28
and VCCO MGB2C 3-29
MGB2C Pin Table 3-30
MGB2C Interconnect Cables and Adapter 3-30

SLR Locations 3-32

Chapter 4: Power

HAPS-200 12F Power States 4-34

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Power State Diagram and Table 4-35
MAINS ON/OFF 4-36
Remote Users 4-36
Power Button LED Behavior 4-36
Power Supply Units 4-37
Battery 4-37

Chapter 5: Clocks

Global Clocks 5-39
Clock Architecture 5-40
Valid Clock Combinations 5-40
MGB2C Clocks 5-42
HC3 Clocks 5-42
Modules vs Number of Clocks Table without Clock Hub 2-10 5-42
Connecting 2 HAPS-200 Modules 5-43
Connecting 6 HAPS-200 12F Modules 5-44
Generating 12 Clocks from a Single HAPS-200 12F Module 5-45
Clock Connectors 5-46
Clock Connector Pin Table 5-47
HC3 Clock Cables 5-47
Universal HC3 Clock Cable (HW1221-0) 5-47
Universal HC3 Loopback Clock Cable (HW1222-0) 5-48
Universal HC3 to HC2 Adapter (HW1238-0) 5-48
Clocking Accessories 5-49
Universal Clock Hub 2-10 (HW1240-0) 5-49
Modules vs Number of Clocks Table with Clock Hub 2-10 5-50

Chapter 6: Getting Started

Special Instructions for Adding Extension Boards and Cables 6-51
Work Flow 6-52
Desktop mode 6-52

Chapter 7: Configuration and Communication

UMRBus® 7-54
Synchronous Reset 7-54
LED Behavior 7-55

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Chapter 8: System

Dimensions 8-56
Weight 8-56

Chapter 9: ProtoCompiler TSS and PCF Examples

PLL Source Clock Example 9-57
Multi-system (6 systems) HAPS-200 12F Board PLL Clock Example 9-59
Mixed System Clocks Example 9-65
Clock Hub Example 9-68
FPGA CLOCKS Examples 9-71
Multisystem without a Clock Hub 9-72
Mixed System Clock Example HAPS-200 12F & HAPS-100 12F 9-74

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CHAPTER 1

General Information

This document is the HAPS-200 12F Reference Manual and describes the functions of the HAPS-200 12F module.

The HAPS ProtoCompiler tools are the dedicated software for HAPS modules. Designed to minimize the effort and time required to bring-up and then deploy a Synopsys® HAPS Series system for IP validation and software development. For more information go to synopsys.com/verification/emulation-prototyping/prototyping.html. ProtoCompiler manuals and user guides can be found under SolvNetPlus > Documentation > ProtoCompiler.

Emissions

The HAPS-200 12F module is a hardware/software development platform that constitutes of an open system design (i.e., has no shielded enclosure) and it has not been tested for compliance with any standard for radio frequency. Since the operation of the product may cause interference with radio communications or with other electronic devices, Synopsys supplies the product to you with the following operational guidance: (a) operate the product only in a shielded laboratory environment, (b) take adequate measures to prevent, reduce or mitigate the effects of undesirable interference, and (c) do not operate the product for use in or with products or applications where the use or malfunction thereof may result in personal injury, death, or catastrophic loss. You assume the entire risk associated with your use of the product outside of these operational guidelines.

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Environmental Guidelines

The conditions recommended in this document apply to the inlet fans of HAPS units with fans or ambient conditions for units without fans.

	Range
Temperature	20 - 25°C (68 - 77°F)
Moisture	40 - 55% Relative Humidity

Ensure Sufficient Airflow

Make sure nothing is blocking the inlet fans in the front or the outlets in the back. For standard desktop use it is recommended to keep a minimum distance of 50 cm between the inlet fans/outlets and any blocking obstacles. When HAPS units are rack-mounted in a data center use a cooling system with a required minimum of 2.5 cm (1 in) free space.

Noise

The maximum noise level with fans running at full speed is 100 dBA.

Support Contact Information

Review this manual as well as other documentation supplied with your HAPS-200 12Fmodule thoroughly before using the equipment. If you require technical support, contact your authorized service technician, or Synopsys directly.

Technical support can be obtained by contacting:

Synopsys Inc.
675 Almanor Avenue
Sunnyvale, CA 94085 USA

solvnetplus.synopsys.com

HAPS Hardware Documentation page on SolvNetPlus has the latest reference manuals, user guides, application notes, board files, firmware, and more.

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IMPORTANT!

ESD



The HAPS-200 12F module, as all other electronic equipment, is sensitive to electrostatic discharge (ESD). Make sure to handle the device properly.

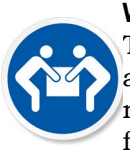
When handling the HAPS-200 12F module:

•Transport the HAPS-200 12F in an ESD case when moving outside an ESD protected environment

•Wear an antistatic wrist strap

•Work area should be ESD secured

Lifting and Moving the HAPS-200 12F Module



WARNING:
The HAPS-200 12F chassis is heavy! Do not attempt to lift the module alone. Use a minimum of two people when lifting the HAPS-200 12F module. Use a server lift to lift and move the HAPS-200 12F. Not following this mandate could result in personal injury and/or damage to the HAPS-200 12F module.

Server Rack Requirements

To prevent injury or damage to the HAPS-200 12F the server rack must have anti-tip measures in place, such as floor-bolting, anti-tip feet, ballast, or a combination as specified by the rack manufacturer and applicable codes.

Before installing the HAPS-200 12F module in the rack, be sure to properly understand the limitations of the rack. You must fully understand the static and dynamic load carrying capacity of the rack and be sure that it can accommodate the weight of the HAPS-200 12F.

Be sure that sufficient clearance exists for cabling, installation and removal of the HAPS-200 12F and movement of the rack doors.

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Technical Specifications

Parameter	HAPS-200 12F
Mains Input Voltage	180-264VAC
Input Frequency	50/60Hz
Input Power	2000W x2
Overvoltage Category	OVC I/II
Indoor/Outdoor Use	Indoor use only
Pollution Degree	PD II
Altitude	Max 2000m (6,562 ft)
Operating Temperature	20 - 25°C (68 - 77°F)
Relative Humidity	40 - 55% non-condensing
Cooling	14,000 BTU/h
Ventilation Requirements	2.5 cm (1 in) minimum free space
Dimensions	800.5 x 448 x 263.7mm (LxWxH) (31.5 x17.6 x 10.4 in)
Weight	75kg (165.4 lbs)

Power Supply

The HAPS-200 12F is included with two CRPS power supplies that allow you to conveniently power your HAPS-200 12F module.

For more information Power, on page 34.

Important Safety Precautions

- Except as described in this manual, all maintenance and repair operations to the HAPS-200 12F power supplies should only be done by a qualified service technician.
- Do not remove the outer cover of the device. Doing so may expose the operator to dangerous voltages. There are no user-serviceable parts inside.
- Do not operate this device if the power cord becomes damaged.
- Only use the cord set provided with your HAPS module. Replacing the cord set with an inadequately rated cord set may result in risk of fire.
- The power cord is considered to be the main disconnect device. To de-energize the equipment, disconnect the power cord.

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- **Pluggable Equipment**: The socket outlet should be installed near the equipment and should be easily accessible. See Properly Grounded Outlet.
- **Class I Equipment**: This equipment must be grounded. The power plug must be connected to a properly wired earth grounded socket outlet. See Properly Grounded Outlet.
- **All Countries**: Install product in accordance with Local and National Electrical Codes.
- If the equipment is not used in a manner specified by the manufacturer, the protection provided by this equipment may be impaired.

Properly Grounded Outlet

The power cord must be plugged into a properly grounded outlet. Failure to have a properly wired outlet may result in damage to the equipment and personal injury. If you are uncertain that suitable grounding is available, consult a licensed electrician to ensure that the electrical outlet is properly grounded.

Hot Plugging is Prohibited

Do not hot plug extension boards and cables. Press the power button for 5 seconds to set the HAPS-200 12F module to OFF before adding or removing extension boards and cables. There is a risk of damaging both the module and extension board if this instruction is not followed.

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Overview

Synopsys® High-Performance ASIC Prototyping System HAPS-200 12F with Symmetrical System Architecture (SSA) offers the best system performance, reliability, and design automation in the industry. The new HAPS-200 series uses the latest AMD Versal™ Premium VP1902 technology. The system has two times the performance and four times debug bandwidth compared to the prior HAPS series. For larger capacity designs the HAPS-200 12F can be expanded by connecting additional HAPS-200 12F, HAPS-200 6F, HAPS-100 4F and HAPS-100 ecosystem.



Figure 1: HAPS-200 12F Module

Features

- 12 AMD XCVP1902-2MSEVSVA6865 User FPGAs
- 336 MG capacity per module; Max 10.8 BG (4 racks)
- 4x Debug bandwidth
- Embedded processor support for user design
- Easy front and back cabling
- 216 HapsTrak 4 and 96 MGB2C
- Selectable I/O voltage (1V - 1.5V) for HapsTrak 4 extension board slots

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- In-system DTD memory buffer with 2x 16GB memory per FPGA
- 32GB DDR available for user FPGA
- 2x 16 GB DDR memory for GSV
- Flexible clock architecture
- UMRBus® 3.0 and ZeBu System Bus support
- Communication and configuration over:
  - USB 3.0
  - 6 QSFP-DD HOST connectors

High level Construction

HAPS-200 12F is a modular design, both in the way it can be used in combination with other products from the HAPS family, but also in the way it has been designed internally.

Access to the HapsTrak 4 and MGB2C connectors is from the Front and Back panels, whereas access to clock and configuration connectors is from the back panel.

The airflow direction for cooling is front-to-back. In other words, cool air is taken in at the front and hot air is exhausted out the back.

FPGA modules and AMD Versal Premium Technology

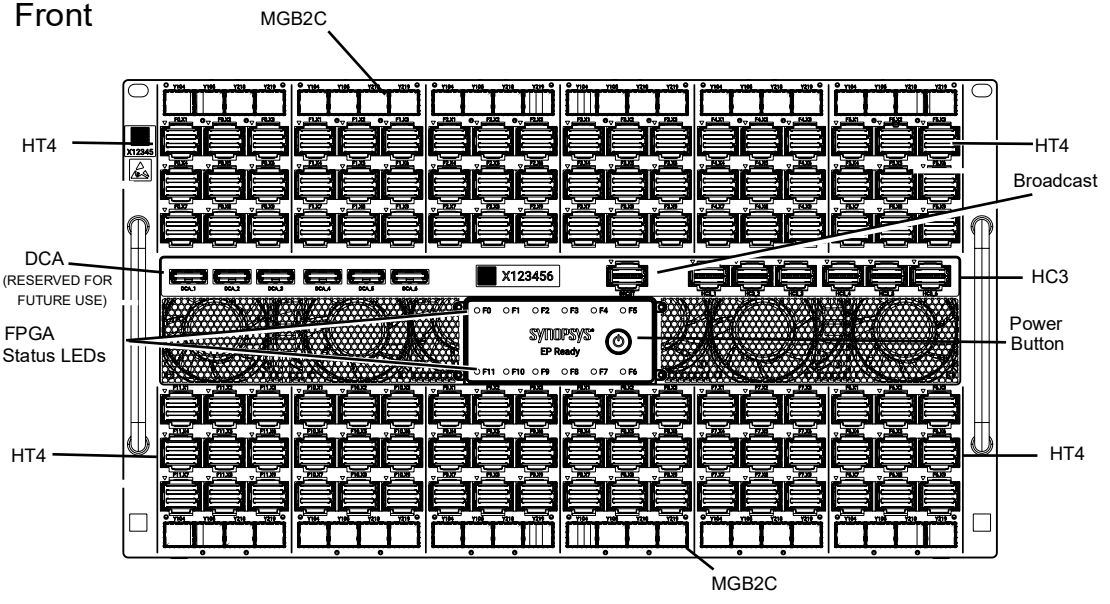
HAPS-200 12F contains twelve FPGA modules each hosting one AMD Versal™ Premium VP1902 (XCVP1902-2MSEVSVA6865) technology device. The adaptive SoC enables scalability with extremely high capacity and connectivity, enhances routability and low latency, and speeds up design iteration.

For more information go to amd.com.

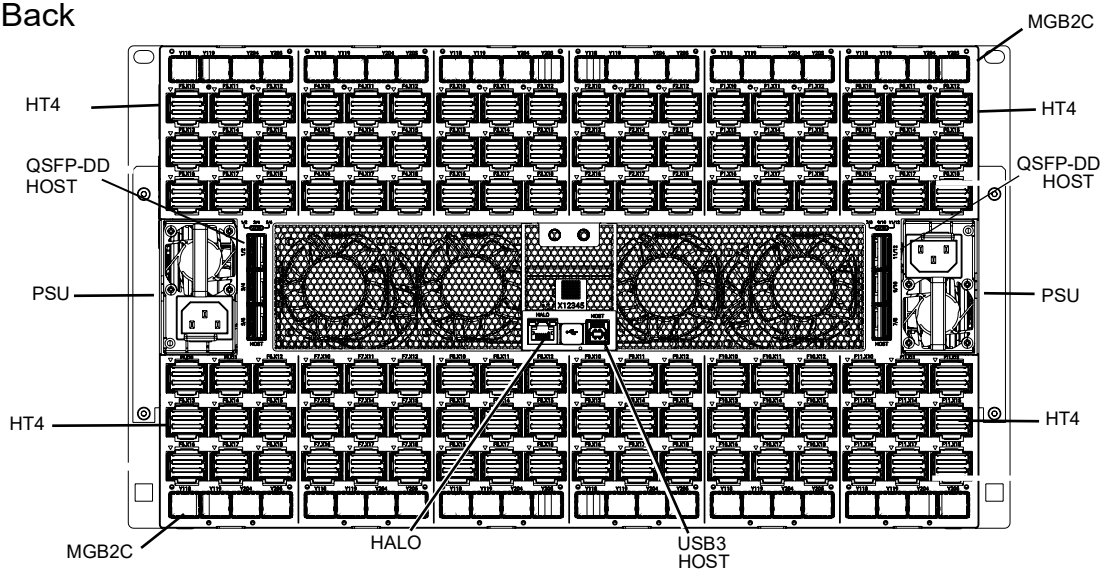
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Layout

Front

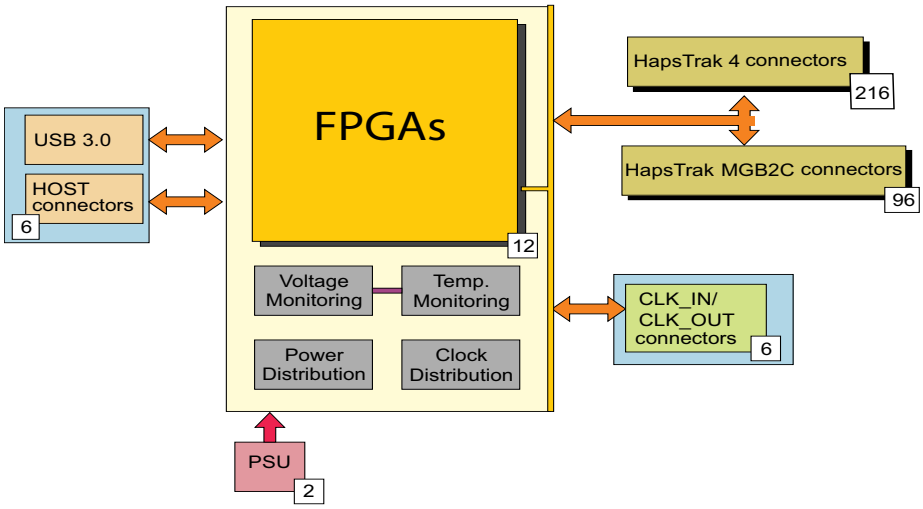


Back



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Block Diagram



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I/O Signals and Connections

On the HAPS-200 12F module FPGAs F0 - F11 have 18 HapsTrak 4 connectors that are extension board and cable capable.

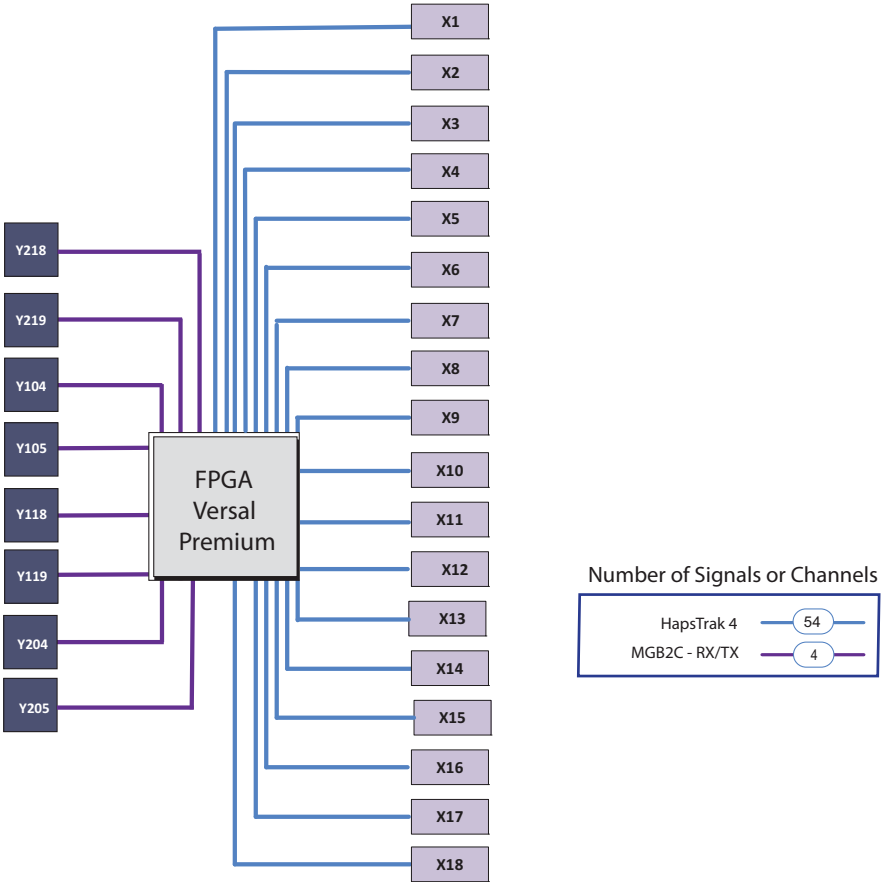


Figure 2: I/O Signals and Connections

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VCCO Regions

The HapsTrak 4 connectors support VCCO voltages 1.0V, 1.1V, 1.2V, 1.35V, and1.5V.

I/O Signals

Each HapsTrak 4 connector on the HAPS-200 12F module has 54 pins used for I/O signaling. See HapsTrak 4 Signals, on page 28.

FPGA Inter-connections

The HAPS-200 12F module has built-in inter-FPGA connections explained in FPGA Inter-connects, on page 23. FPGA inter-connections can also be done through flexible interconnect cables available in different lengths.

System IP and General Purpose I/Os

The System IP is a module that the ProtoCompiler automatically inserts for each HAPS-200 12F FPGA. It consolidates various operations including multiplexing, serialization and de-serialization, local resets, clock de-skewing, UMR chaining, and PCB trace assignment for GPIO. For more information refer to the *ProtoCompiler User Guide*.

LEDs

The module has the following indicators:

- F0-F11 LEDs - FPGA status indicators, see LED Behavior, on page 55.
- HOST connector status LEDs - Indicate HOST connector status. See Configuration and Communication, on page 54.
- POWER button LED - Indicates power status. See Power Button LED Behavior, on page 36.

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Clocks

A HAPS-200 12F module has 24 global clocks (16 XPIO and 8 HDIO). All global clocks can be used as independent clocks in each FPGA or shared across FPGAs. Global clocks can be shared between 6 HAPS-200 12F/HAPS-200 6F modules without using a Clock Hub.

For more information see Clocks, on page 38.

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CHAPTER 2

Concept

HapsTrak

HAPS is a system solution consisting of software and hardware that aims to provide the most productive and efficient way of doing FPGA prototyping.

Important features that make HAPS a powerful FPGA prototyping system are:

- Access to FPGA I/O signals through the proven HapsTrak 4 and MGB2C connectors
- Highly flexible ways of creating interconnect structures through built-in interconnects, cables and interconnect cards
- Extension boards and interface cards connect to the HapsTrak 4 or MGB2C connectors and provide access to, for example, DDR memory, FLASH, and PCIe connections
- Convenient and scalable ways to connect multiple HAPS systems in cloud configurations

HapsTrak is a mechanical and electrical standard for attaching extension boards and interconnects to a HAPS prototyping system. It specifies mechanical dimensions, connector types and locations, as well as pin out, electrical signal properties, and power rules. To be HAPS compliant, all systems, motherboards, and extension boards must follow these standards.

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HapsTrak 4

The HapsTrak 4 (HT4) is a 150-pin connector containing 60 signal pins, that can be used for single ended or differential signaling. HAPS-200 12F uses 54 signal pins, which are available in the created bus. See HapsTrak 4 Signals, on page 28.

All signals in a connector are associated with the same FPGA I/O bank voltage (VCCO). The I/O voltage dictates which electrical I/O standards can be used for the signal pins.

HapsTrak MGB2C

The HAPS MGB (Multi-GigaBit) family is a collection of interface cards and cables, built around the HapsTrak MGB bus and supported by HAPS systems.

MGB2C is the second adaptation of the MGB2 standard. MGB2C is based on Samtecs Novaray connector family and is rated to 56 Gbps NRZ/112 Gbps PAM4. The speed achieved depends on internal and external cable lengths. The MGB2C connector does not support extension boards without an MGB2C to MGB2 adapter. MGB2 is the only connector in the MGB2 family that supports extension boards.

For more information concerning the MGB2C connector, see and VCCO MGB2C, on page 29. For more information about the available cards and adapters, go to solvnetplus.synopsys.com.

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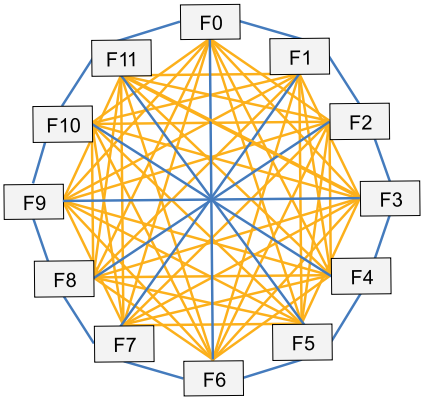
FPGA Inter-connects

GTYP

HAPS-200 12F has fixed inter-FPGA connections between the 12 FPGAs. Every FPGA will connect to every other FPGA with one or two half-transceiver quads. The internal interconnect is made so that every FPGA connects to another FPGA in with the same bank mapping. This allows you to place a design that uses, for example, F0, F1 and F2 in any other three adjacent FPGAs. The ability to move designs is called ***relocatablity***, for more information go to ProtoCompiler Reference Manual and User Guide.

GT Transceiver Half Quads

	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
F0		2	1	1	1	1	2	1	1	1	1	2
F1	2		2	1	1	1	1	2	1	1	1	1
F2	1	2		2	1	1	1	1	2	1	1	1
F3	1	1	2		2	1	1	1	1	2	1	1
F4	1	1	1	2		2	1	1	1	1	2	1
F5	1	1	1	1	2		2	1	1	1	1	2
F6	2	1	1	1	1	2		2	1	1	1	1
F7	1	2	1	1	1	1	2		2	1	1	1
F8	1	1	2	1	1	1	1	2		2	1	1
F9	1	1	1	2	1	1	1	1	2		2	1
F10	1	1	1	1	2	1	1	1	1	2		2
F11	2	1	1	1	1	2	1	1	1	1	2	



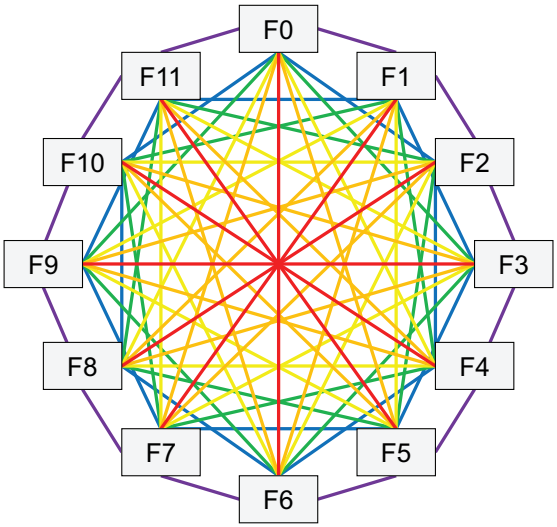
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XPIO

Like GTYP interconnect, the XPIO internal interconnect is made so that every FPGA connects to another FPGA with the same bank mapping to allow you to place a design that, for example, uses FO, F1, and F2 in any other three adjacent FPGAs.

XPIO Bank Interconnection

	F0	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11
F0		2	2	1	1	1	1	1	1	1	2	2
F1	2		2	2	1	1	1	1	1	1	1	2
F2	2	2		2	2	1	1	1	1	1	1	1
F3	1	2	2		2	2	1	1	1	1	1	1
F4	1	1	2	2		2	2	1	1	1	1	1
F5	1	1	1	2	2		2	2	1	1	1	1
F6	1	1	1	1	2	2		2	2	1	1	1
F7	1	1	1	1	1	2	2		2	2	1	1
F8	1	1	1	1	1	1	2	2		2	2	1
F9	1	1	1	1	1	1	1	2	2		2	2
F10	2	1	1	1	1	1	1	1	2	2		2
F11	2	2	1	1	1	1	1	1	1	2	2	



Interconnect Boards and Cables

On HAPS systems the majority of I/O signals are available on HapsTrak connectors. The connectors can be used either to add external logic or for connectivity between FPGAs.

For more information see the *Interconnect Boards and Cables Reference Manual* on HAPS SupportNet.

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Extension Boards

Each extension board connects with one or several connectors on the HAPS system. Extension boards can be moved around freely, allowing reconfiguration of the prototype system as parts of the design moves from one FPGA to another.

An extension board for HAPS-200 12F has AcceleRate®HP HapsTrak 4 connectors, and typically contains peripheral or interconnect circuitry. Extension boards are oriented horizontally, and can be designed to span one or several connectors.

There is also a Universal HT4 to HT3 Adaptor that maps HT3 type extension boards to HT4 standard.

Important! Make sure that extension boards that extend outside of the HAPS-200 12F system do not interfere with neighboring systems or other hardware.

For making custom extension boards refer to *Extension Board Design Guide* found at HAPS Hardware Documentation on SolvNetPlus.

Multi-Module Setups

HAPS-200 12F modules have been designed to support prototyping needs ranging from single module up to large network connected setups. Furthermore, HAPS-200 12F modules can function together with HAPS-200 and HAPS-100 systems. This creates a flexible and powerful environment where subcomponents of a large design can be verified independently using smaller HAPS setups, while still allowing a full scale design verification in the larger setup.

Synopsys offers mechanical equipment for HAPS-200 12F rack use. Contact Synopsys Support for more information.

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Available DDR4 Memory

In-system DTD Memory Buffer - Each user FPGA includes two dedicated 16 GB DDR4 SODIMMs used by the in-system DTD memory buffer for debug and data capture functions.

DDR4 available for user FPGA - Each user FPGA has an additional 32 GB fabric-facing DDR4 memory accessible to the programmable logic for user designs.

DDR4 memory for GSV - Each user FPGA is connected to two 16 GB DDR4 SODIMMs through the NoC, providing memory resources for the GSV debug and User PS subsystem. For more information see the *ProtoCompiler User Guide*.

For DDR4 SLR locations, see SLR Locations, on page 32 and SLR Connectors, on page 33.

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CHAPTER 3

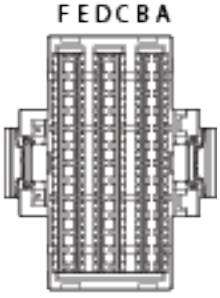
Expansion and I/Os

The connectors on HAPS-200 12F are used for expansion of a HAPS system with extension boards or cables, either supplied by Synopsys, HAPS Connect, or custom-made. The I/O connectors comply with the HapsTrak Standard which can be downloaded from *HAPS Hardware Documentation Page* on *SolvNet-Plus*.

I/O Connectors

HapsTrak 4

The HT4 connectors are located on the front and back sides of the HAPS-200 12F system. A total of 216 HT4 connectors are available. Each of the 12 FPGAs has 18 HT4 connectors that are extension board capable.



The HT4 connectors for all 4 SLRs are mechanically arranged consecutive to each other. This is important for extension boards that require multiple connectors in the same SLR.

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VCCO

The HT4 connectors support VCCO voltages 1.5V, 1.35V, 1.2V, 1.1V, and 1.0V.

HapsTrak 4 Signals

The HT4 connector definitions for HAPS-200 12F are listed in the following pin diagram.

	A	B	C	D	E	F
1	GND	GND	GND	GND	GND	GND
2	A0	B0	12V_IN	12V_OUT	E0	F0
3	GND	GND	GND	GND	GND	GND
4	A1	B1	SCL_IN	SCL_OUT	E1	F1
5	GND	GND	GND	GND	GND	GND
6	A2	B2	SDA_IN	SDA_OUT	E2	F2
7	GND	GND	GND	GND	GND	GND
8	A3	B3	C0	D0	E3	F3
9	GND	GND	GND	GND	GND	GND
10	A4	B4	C1	D1	E4	F4
11	GND	GND	GND	GND	GND	GND
12	A5	B5	C2	D2	E5	F5
13	GND	GND	GND	GND	GND	GND
14	A6	B6	C3	D3	E6	F6
15	GND	GND	GND	GND	GND	GND
16	A7	B7	C4	D4	E7	F7
17	GND	GND	GND	GND	GND	GND
18	A8	B8	C5	D5	E8	F8
19	GND	GND	GND	GND	GND	GND
20	A9	B9	Id0	Id1	E9	F9
21	GND	GND	GND	GND	GND	GND
22	A10	B10	Id2	Id3	E10	F10
23	GND	GND	GND	GND	GND	GND
24	A11	B11	VREF	Id4	E11	F11
25	GND	GND	GND	GND	GND	GND

Legend

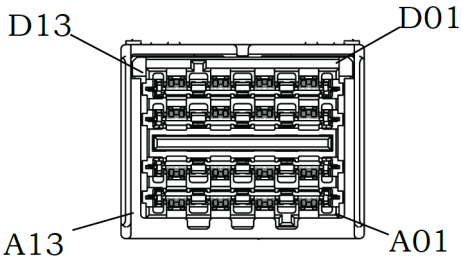
GND	Ground
SCL_IN	I2C controller in remote module (ID PROM in local module)
SDA_IN	
SCL_OUT	I2C controller in local module (ID PROM in remote module)
SDA_OUT	
12V_IN	12V supplied by a remote module
12V_OUT	12V supplied by the local module
VREF	Reference voltage supplied by the local module
xx	I/O
xx	Rx Strobe
Idx	Identification pins clock, data, and address

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and VCCO **MGB2C**

HAPS-200 12F has 8 (GTYP) MGB2C connectors per FPGA located in the Front and Back sides, see Layout, on page 16. MGB2C connectors have higher performance than prior HAPS system's MGB2 connectors.

-



FPGA	Type	MGB2C Type GTYP Connectors
F0	Front	F0_Y104, F0_Y105, F0_Y218, F0_Y219
	Back	F0_Y118, F0_Y119, F0_Y204, F0_Y205
F1	Front	F1_Y104, F1_Y105, F1_Y218, F1_Y219
	Back	F1_Y118, F1_Y119, F1_Y204, F1_Y205
F2	Front	F2_Y104, F2_Y105, F2_Y218, F2_Y219
	Back	F2_Y118, F2_Y119, F2_Y204, F2_Y205
F3	Front	F3_Y104, F3_Y105, F3_Y218, F3_Y219
	Back	F3_Y118, F3_Y119, F3_Y204, F3_Y205
F4	Front	F4_Y104, F4_Y105, F4_Y218, F4_Y219
	Back	F4_Y118, F4_Y119, F4_Y204, F4_Y205
F5	Front	F5_Y104, F5_Y105, F5_Y218, F5_Y219
	Back	F5_Y118, F5_Y119, F5_Y204, F5_Y205
F6	Front	F6_Y104, F6_Y105, F6_Y218, F6_Y219
	Back	F6_Y118, F6_Y119, F6_Y204, F6_Y205
F7	Front	F7_Y104, F7_Y105, F7_Y218, F7_Y219
	Back	F7_Y118, F7_Y119, F7_Y204, F7_Y205
F8	Front	F8_Y104, F8_Y105, F8_Y218, F8_Y219
	Back	F8_Y118, F8_Y119, F8_Y204, F8_Y205
F9	Front	F9_Y104, F9_Y105, F9_Y218, F9_Y219
	Back	F9_Y118, F9_Y119, F9_Y204, F9_Y205
F10	Front	F10_Y104, F10_Y105, F10_Y218, F10_Y219
	Back	F10_Y118, F10_Y119, F10_Y204, F10_Y205
F11	Front	F11_Y104, F11_Y105, F11_Y218, F11_Y219
	Back	F11_Y118, F11_Y119, F11_Y204, F11_Y205

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For guidance in designing application specific MGB2 interface cards, consult the MGB2 Interface Card Design Guide on SolvNetPlus HAPS Hardware Documentation page.

MGB2C Pin Table

13	12	11	10	9	8	7	6	5	4	3	2	1	
GND	SDA_OUT	GPIO3	GND	TX1n	TX1p	GND	REFCLK0N	REFCLK0P	GND	TX0n	TX0p	GND	D
GND	12V_OUT	GPIO1	GND	TX3n	TX3p	GND	SCL_OUT	GPIO2	GND	TX2n	TX2p	GND	C
GND	12V_IN	GPIO6	GND	RX3n	RX3p	GND	SCL_IN	GPIO5	GND	RX2n	RX2p	GND	B
GND	SDA_IN	GPIO4	GND	RX1n	RX1p	GND	REFCLK1N	REFCLK1P	GND	RX0n	RX0p	GND	A

MGB2C Interconnect Cables and Adapter

Cables are available to connect MGB2C to MGB2 and MGB2B. For more information see Interconnect Boards & Cables Reference Manual on SolvNetPlus HAPS Hardware Documentation page.

MGB2C to MGB2C Interconnect Cables

MGB2C Standard MGB2C interconnect cables for connections between MGB2C connectors.

The following is a table of the available cables with the HW numbers and lengths.

HW#	Length (cm)
HW1261-0	25
HW1223-0	50
HW1224-0	100
HW1225-0	200
HW1226-0	300
HW1227-0	400
HW1228-0	500



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MGB2C to MGB2 Interconnect Cables

MGB2C Standard MGB2 interconnect cables for connections between an MGB2C connector and an MGB2 connector.

HW#	Length (cm)
HW1229-0	25
HW1230-0	50
HW1231-0	100
HW1232-0	200
HW1233-0	300
HW1234-0	400
HW1235-0	500



MGB2C to MGB2B Interconnect Cables

MGB2C Standard MGB2B interconnect cables for connections between an MGB2C connector and an MGB2B connector.

HW#	Length (cm)
HW1267-0	50
HW1268-0	100
HW1269-0	150
HW1270-0	200
HW1271-0	300
HW1272-0	400
HW1273-0	500



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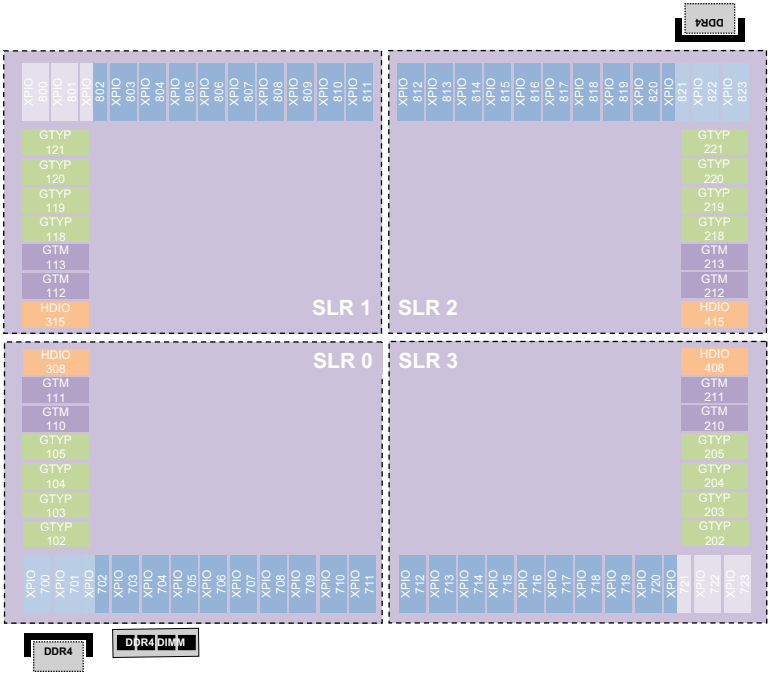
SLR Locations

Respect SLR (Super Logic Region) locations for interconnects and minimize routing between SLRs inside the FPGA. Respecting SLR locations generally renders logic with less clock skew within local region user logic. Certain restrictions exist in regard to clock resource use between SLRs. For more information go to amd.com.

Avoid crossing SLR borders with an extension board that has multiple HapsTrak 4 connectors and sources synchronous signals to the FPGA.

AMD VP1902 FPGA SLRs are arranged in a 2x2 matrix. SLR0 and SLR2 are diagonally opposite and SLR1 and SLR3 diagonally opposite.

HAPS-200 12F connectors are positioned so that cabling from front to back is minimized and SLR can be reached from either front or back, in most cases, with a single SLR jump.



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SLR Connectors

	SLR	Bank	Connector	Type
FRONT	0	709	X1	HT4
	0	710	X2	HT4
	0	711	X3	HT4
	0	708	X4	HT4
	2	816	X5	HT4
	2	815	X6	HT4
	2	814	X7	HT4
	2	813	X8	HT4
	2	812	X9	HT4
BACK	1	809	X10	HT4
	1	810	X11	HT4
	1	811	X12	HT4
	1	808	X13	HT4
	3	716	X14	HT4
	3	715	X15	HT4
	3	714	X16	HT4
	3	713	X17	HT4
	3	712	X18	HT4
FRONT	0	104	Y104	MGB2C
	0	105	Y105	MGB2C
	2	218	Y218	MGB2C
	2	219	Y218	MGB2C
BACK	1	118	Y118	MGB2C
	1	119	Y119	MGB2C
	3	204	Y204	MGB2C
	3	205	Y205	MGB2C
INTERNAL	0	700		DDR4-NOC
	0	701		DDR4-NOC
	0	702	Half Bank	DDR4-NOC
	0	703		DDR4-DIMM
	0	704		DDR4-DIMM
	0	705		DDR4-DIMM
	2	821	Half Bank	DDR4-NOC
	2	822		DDR4-NOC
	2	823		DDR4-NOC

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CHAPTER 4

Power

HAPS-200 12F Power States

The HAPS-200 12F has three power states, OFF, STANDBY, and ON. These states are defined in the firmware and subject to change. Always check that you have the latest firmware and check which features are included in each power state.

OFF

The HAPS-200 12F must be in the OFF state when repopulating extension boards or adding/changing HC3, HT4, MGB2C, or DCA cables. The module must also be in the OFF state before switching off MAINS.

Multi-Design Mode (MDM) is supported, but the entire HAPS-200 12F must be in the OFF state for changing cables/extension boards.

STANDBY

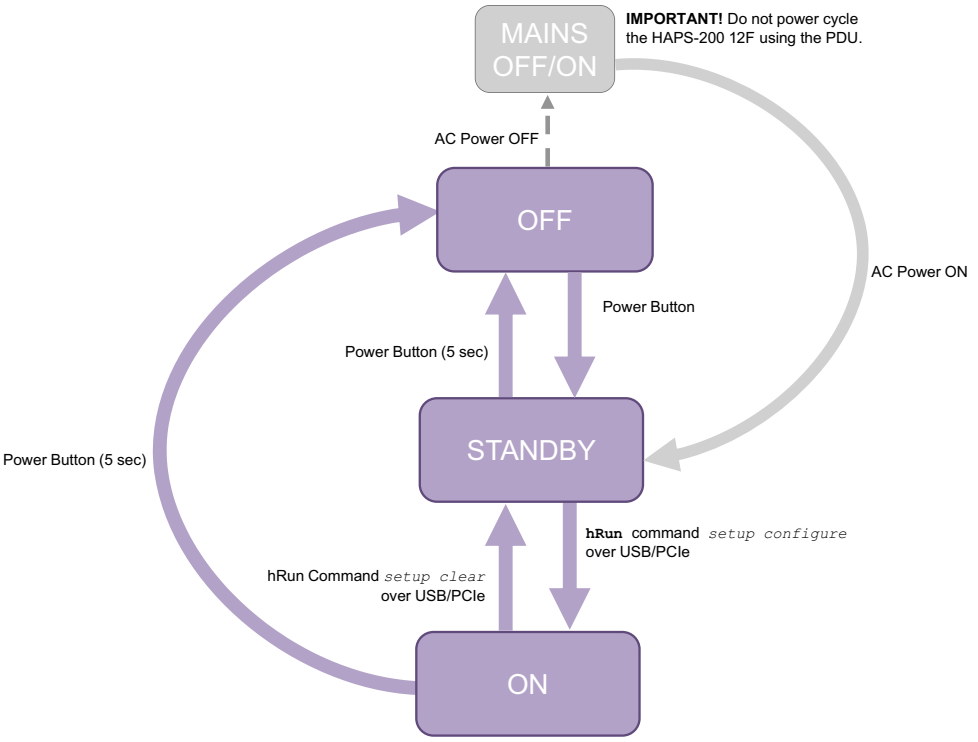
This state is reached after first Power ON. See Power State Diagram and Table, on page 35.

ON

This state is reached after at least one FPGA is configured.

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Power State Diagram and Table



Current Power State	Transition to Next State		HT4 12V	HT4 VCCO (VREF)
OFF	Press Power Button	STANDBY	NO	NO
STANDBY	*setup configure*	ON	NO	NO
	Press Power Button (5 sec)	OFF		
ON	*setup clear*	STANDBY	YES*	YES*
	Press Power Button (5 sec)	OFF		

*If the connector is used.

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MAINS ON/OFF

The HAPS-200 12F must be in the OFF state before switching off MAINS. Not following this mandate may cause long term stress, reliability and malfunction issues in the HAPS-200 12F module.

IMPORTANT! Do not use the PDU to power cycle the HAPS-200 12F.

Remote Users

USB HOST
: Can power ON the system from OFF state and power cycle the system.

PCIe HOST
: Can power cycle only.

Power Button LED Behavior

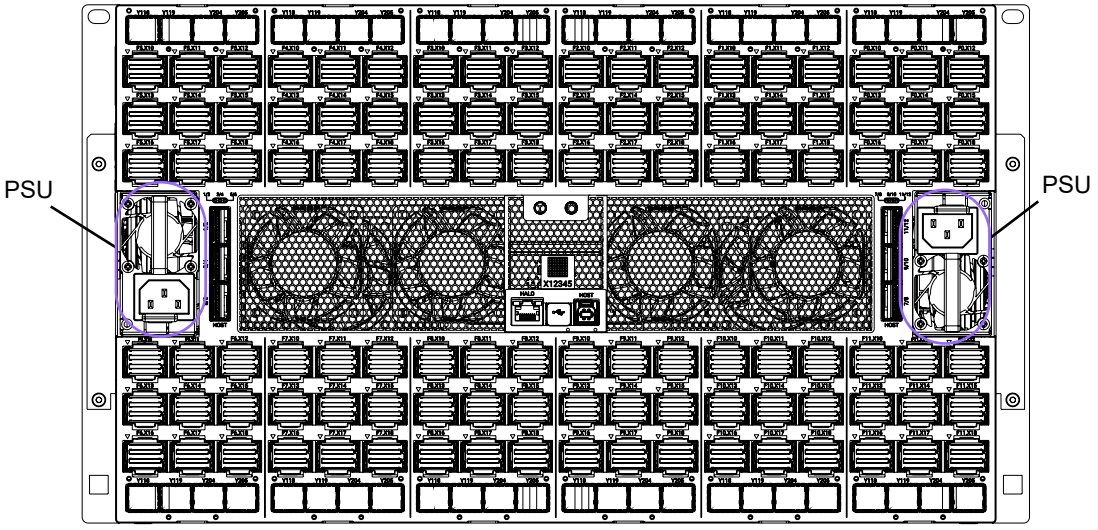
LED Color	Behavior	State
Off	Power not active	OFF
Amber	12 V power on/ Module power state OFF	
Green flashing	12 V power on/ Booting or powering down	
Green	Ready to use. Module power state is STANDBY.	STANDBY/ ON
Green/Amber flashing	Handle open by at least one host	
Red flashing	Error (for example, fallback design running)	
Red	Severe error (for example, power error)	

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Power Supply Units

The HAPS-200 12F module is powered by two internal PSUs (2KW each) used in parallel. The PSUs have C14 input power connectors.

.



Battery

Batteries are installed inside the HAPS-200 12F for encryption and realtime clock usage.

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CHAPTER 5

Clocks

The HAPS-200 clock distribution is similar to prior HAPS modules.

Clocks can source from:

- Onboard PLL
- FPGA
- From another HAPS-200 module
  - For more than 6 modules a clock hub is needed, see Modules vs Number of Clocks Table with Clock Hub 2-10, on page 50
- From a HAPS-100 module
  - A clock hub is needed, see Universal Clock Hub 2-10 (HW1240-0), on page 49.

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Global Clocks

24 global clocks (GC or GCLK) are connected to clock capable pins. Global clocks GCLK3, GCLK6, GCLK9, GCLK12, GCLK15, GCLK18, GCLK21, and GCLK24 are connected to HDIO banks.

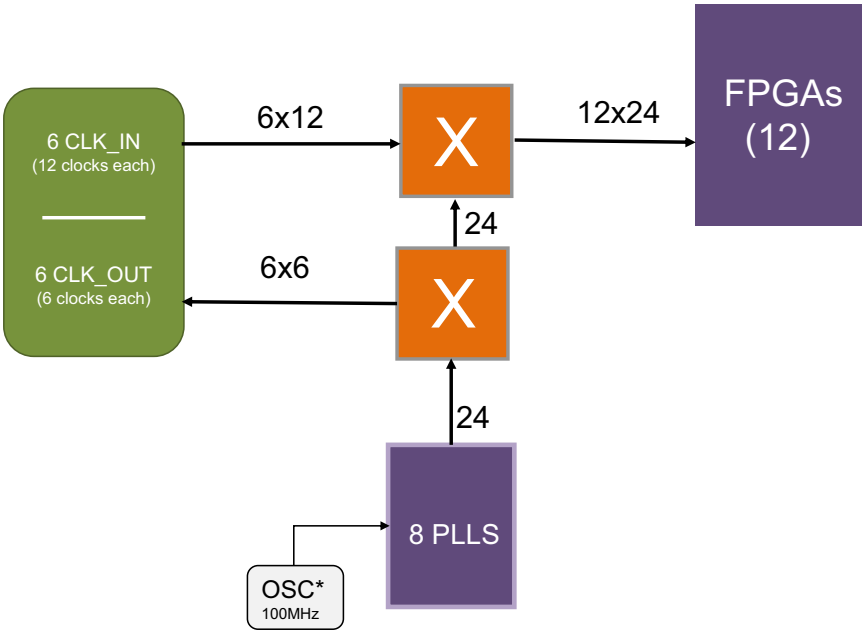
All global clocks can be used as independent clocks in each FPGA or shared across FPGAs. Global clocks can be shared between 6 HAPS-200 modules without requiring a clock hub.

Primary port-level synchronous clocks must be pin-locked within the same I/O bank. The table below shows the bank locations for the 24 global clocks. The color coding indicates which clocks share the same user FPGA bank.

Clock	Bank	SLR
GCLK1	702	0
GCLK2	702	
GCLK3	308	
GCLK4	802	1
GCLK5	802	
GCLK6	315	
GCLK7	821	2
GCLK8	821	
GCLK9	415	
GCLK10	721	3
GCLK11	721	
GCLK12	408	
GCLK13	703	0
GCLK14	704	
GCLK15	308	
GCLK16	803	1
GCLK17	804	
GCLK18	315	
GCLK19	820	2
GCLK20	819	
GCLK21	415	
GCLK22	720	3
GCLK23	719	
GCLK24	408	

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Clock Architecture



*Each PLL has a separate oscillator input enabling all 8 PLLs to run on independent clocks. See PLL Clocks, on page 40 for more information.

PLL Clocks

One oscillator, also used for GCLK0 distribution, generates the input clocks for all PLLs. All the generated clocks are common source and do not drift from each other. The clocks are phase locked and the phase shift is unknown.

Valid Clock Combinations

Global clocks 1,4,7,10,13,16,19 and 22 of each FPGA can be connected to CLK1 output of any of the eight available PLLs. Valid combinations for the remaining PLL clocks are shown in the below table.

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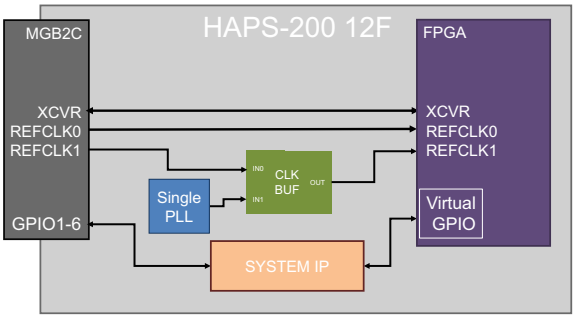
The CLK_IN1 pin of any six available clock connectors can source to FPGA GCLK1 and GCLK7. Valid combinations from CLK_IN pins to GCLKs are shown in the table below.

Clock	Sinks	Valid Sources
FPGA	F[0-11].CLK1,7	PLL[1-8].CLK1
		HC3[1-6].CLKIN1
		SRC1|SRC2|SRC3|SRC4
	F[0-11].CLK2,8	PLL[1-8].CLK2
		HC3[1-6].CLKIN2
		SRC5|SRC6|SRC7|SRC8
	F[0-11].CLK3,9	PLL[1-8].CLK3
		HC3[1-6].CLKIN3
		SRC9|SRC10|SRC11|SRC12
	F[0-11].CLK4,10	PLL[1-8].CLK1
		HC3[1-6].CLKIN4
		SRC1|SRC2|SRC3|SRC4
	F[0-11].CLK5,11	PLL[1-8].CLK2
		HC3[1-6].CLKIN5
		SRC5|SRC6|SRC7|SRC8
	F[0-11].CLK6,12	PLL[1-8].CLK3
		HC3[1-6].CLKIN6
		SRC9|SRC10|SRC11|SRC12
	F[0-11].CLK13,19	PLL[1-8].CLK1
		HC3[1-6].CLKIN7
		SRC1|SRC2|SRC3|SRC4
	F[0-11].CLK14,20	PLL[1-8].CLK2
		HC3[1-6].CLKIN8
		SRC5|SRC6|SRC7|SRC8
	F[0-11].CLK15,21	PLL[1-8].CLK3
		HC3[1-6].CLKIN9
		SRC9|SRC10|SRC11|SRC12
	F[0-11].CLK16,22	PLL[1-8].CLK1
		HC3[1-6].CLKIN10
		SRC1|SRC2|SRC3|SRC4
	F[0-11].CLK17,23	PLL[1-8].CLK2
		HC3[1-6].CLKIN11
		SRC5|SRC6|SRC7|SRC8
	F[0-11].CLK18,24	PLL[1-8].CLK3
		HC3[1-6].CLKIN12
		SRC9|SRC10|SRC11|SRC12

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MGB2C Clocks

Each MGB2C quad will get REFCLK0 directly from the connector, REFCLK1 from the connector, or a programmable PLL.



HC3 Clocks

Only CLK_IN[1..6] in the HC3 connector can be used when connecting two HAPS-200s with an HC3 Cable. CLK_IN[7..12] (as well as CLK_IN[1..6]) can be used when connecting multiple HAPS-200 modules through a clock hub.

Modules vs Number of Clocks Table without Clock Hub 2-10

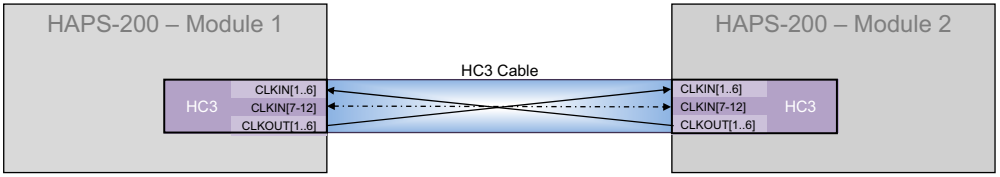
The following table shows the maximum number of possible global clocks per the number of modules.

Number of Modules	Maximum Number of Possible Global Clocks
1	24
2	12
3	12

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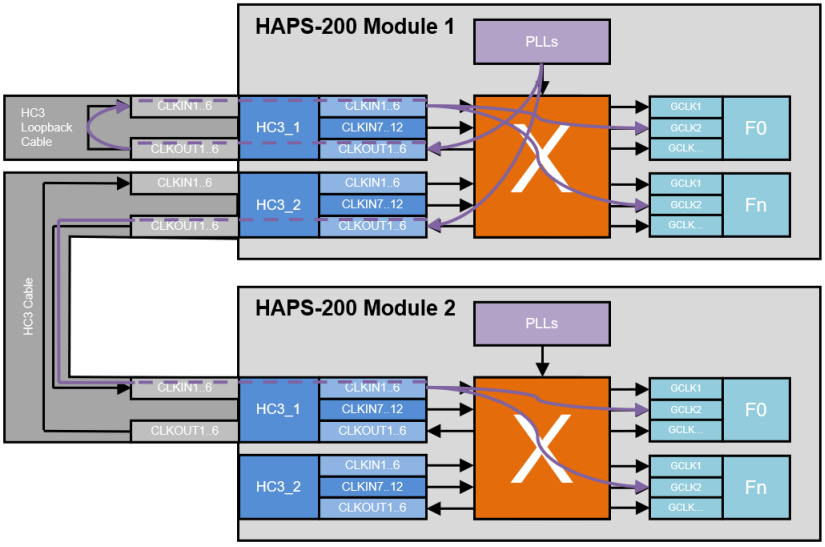
Connecting 2 HAPS-200 Modules

Connecting clock connectors from one HAPS-200 module to another with an HC3 cable. CLK_OUT[1..6] from the first module cross-connects to CLK_IN[1..6] of the second module.



The following figure illustrates connecting two HAPS-200 modules with an HC3 Loopback Cable and a standard HC3 Cable.

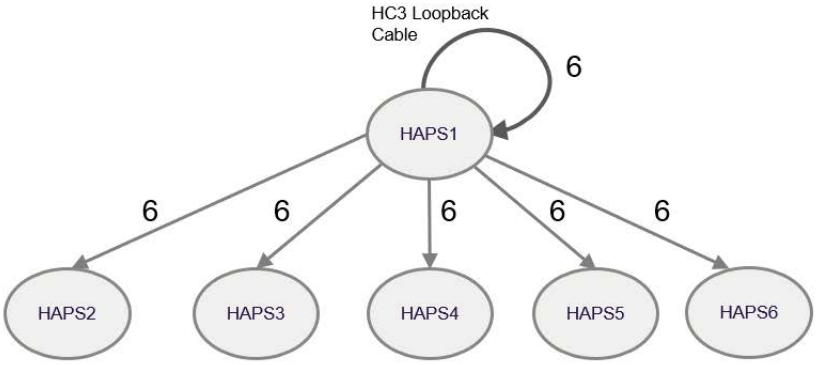
- Module 1 generates a PLL clock and sends it to HC3_1 and HC3_2.
- Using an HC3 Loopback cable the Module 1 PLL clock is sent back to Module 1 HC3_1 and further to all Fx GCLK2 in Module 1.
- Module 1 HC3_2 sends the clock to Module 2 HC3_1 and further to all Fx GCLK2 in Module 2.



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Connecting 6 HAPS-200 12F Modules

An HC3 loopback cable is needed for self- synchronization on the clock-generating HAPS-200 module. 6 clocks can be distributed to 6 (1+5) HAPS-200 modules using HC3 cables without using a clock hub. One HAPS-200 12F module generates clocks and uses an HC3 connector to loop the clocks back, the five remaining HC3 connectors can go to five other modules

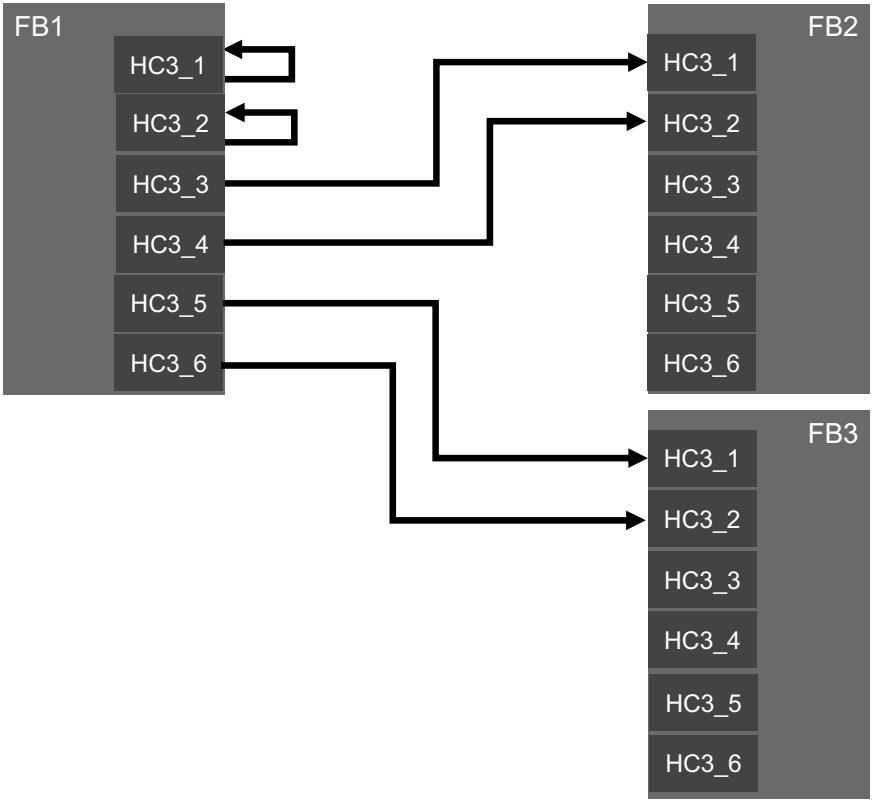


For TSS examples see ProtoCompiler TSS and PCF Examples, on page 57.

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Generating 12 Clocks from a Single HAPS-200 12F Module

Twelve clocks can be generated from a single HAPS-200 12F module to itself and two other HAPS-200 12F Modules.

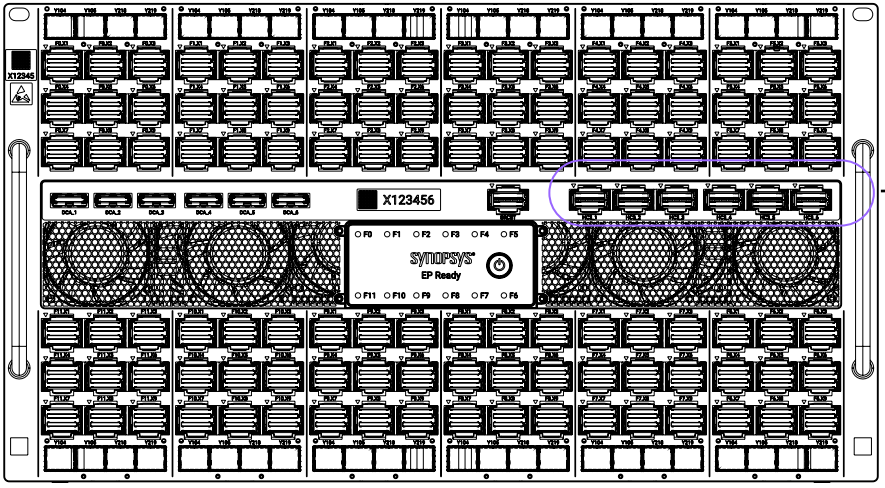


For connection examples and TSS/PCF information, see ProtoCompiler TSS and PCF Examples, on page 57.

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Clock Connectors

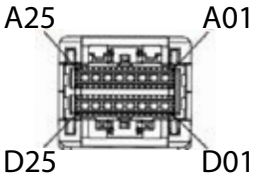
HC3 clock connectors are used to connect clocks between HAPS-200 modules. There are 6 connectors with CLK_IN and CLK_OUT bundled in one connector. There are 12 CLK_IN and 6 CLK_OUT per connector.



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Clock Connector Pin Table

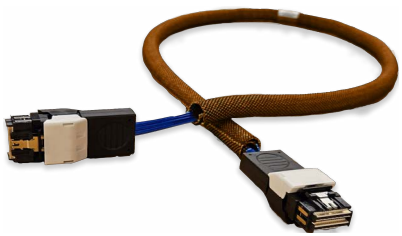
	A	B	C	D
1	GND	GND	GND	GND
2	CLK_IN1_P	12V_IN	12V_OUT	CLK_OUT1_P
3	GND	GND	GND	GND
4	CLK_IN1_N	SCL_IN	SCL_OUT	CLK_OUT1_N
5	GND	GND	GND	GND
6	CLK_IN2_P	SDA_IN	SDA_OUT	CLK_OUT2_P
7	GND	GND	GND	GND
8	CLK_IN2_N	CLK_IN7_P	CLK_IN10_P	CLK_OUT2_N
9	GND	GND	GND	GND
10	CLK_IN3_P	CLK_IN7_N	CLK_IN10_N	CLK_OUT3_P
11	GND	GND	GND	GND
12	CLK_IN3_N	CLK_IN8_P	CLK_IN11_P	CLK_OUT3_N
13	GND	GND	GND	GND
14	CLK_IN4_P	CLK_IN8_N	CLK_IN11_N	CLK_OUT4_P
15	GND	GND	GND	GND
16	CLK_IN4_N	CLK_IN9_P	CLK_IN12_P	CLK_OUT4_N
17	GND	GND	GND	GND
18	CLK_IN5_P	CLK_IN9_N	CLK_IN12_N	CLK_OUT5_P
19	GND	GND	GND	GND
20	CLK_IN5_N	Id0	Id1	CLK_OUT5_N
21	GND	GND	GND	GND
22	CLK_IN6_P	Id2	Id3	CLK_OUT6_P
23	GND	GND	GND	GND
24	CLK_IN6_N	VREF	Id4	CLK_OUT6_N
25	GND	GND	GND	GND



HC3 Clock Cables

Universal HC3 Clock Cable (HW1221-0)

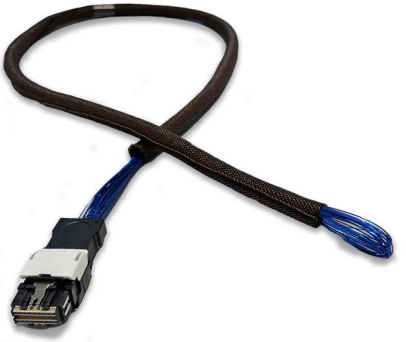
The clock cables allow for in-system identification of the clock source. Below is a picture of one of the many HC3 clock cables available.



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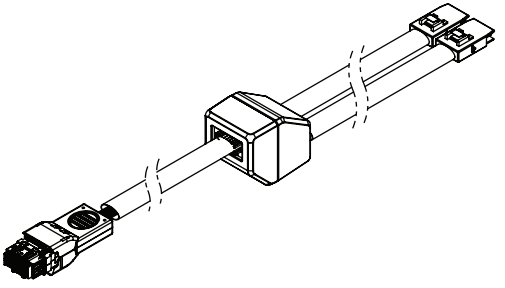
Universal HC3 Loopback Clock Cable (HW1222-0)

An HC3 Loopback Cable is needed on the clock-generating HAPS-200 module for self-synchronization.



For information concerning the HC3 and the HC3 Loopback cables, see the *Interconnect Boards & Cables*.

Universal HC3 to HC2 Adapter (HW1238-0)



The Universal HC3 to HC2 Adapter (HC2X2_ADAPTER_HC3) synchronizes clocks between HAPS-200 and HAPS-100 modules directly or through a Clock Hub, for example, Clock HUB 2-10. The adapter plugs directly into the HAPS-200 12F HC3 connector and divides the HC3 connector 6 input clocks into 2 HC2 connectors with 6 clocks each. If connected to a Clock Hub 2-10, 12 output clocks from HC3 feeds 12 clocks (6+6) on HC2 side.

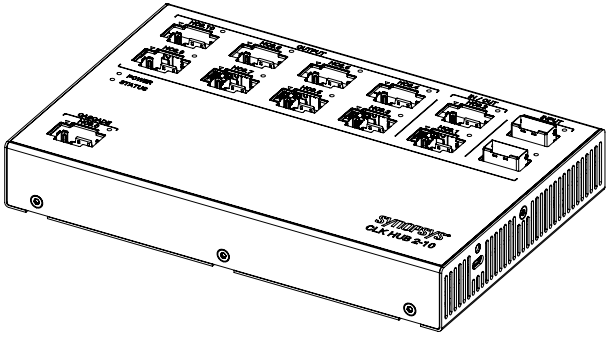
The adapter has an HC3 connector with a factory-installed HC3 91.4 cm cable and two factory-installed universal HC2 48 cm cables.

For more information see *Universal HC3 to HC2 Adapter Reference Manual*.

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Clocking Accessories

Universal Clock Hub 2-10 (HW1240-0)



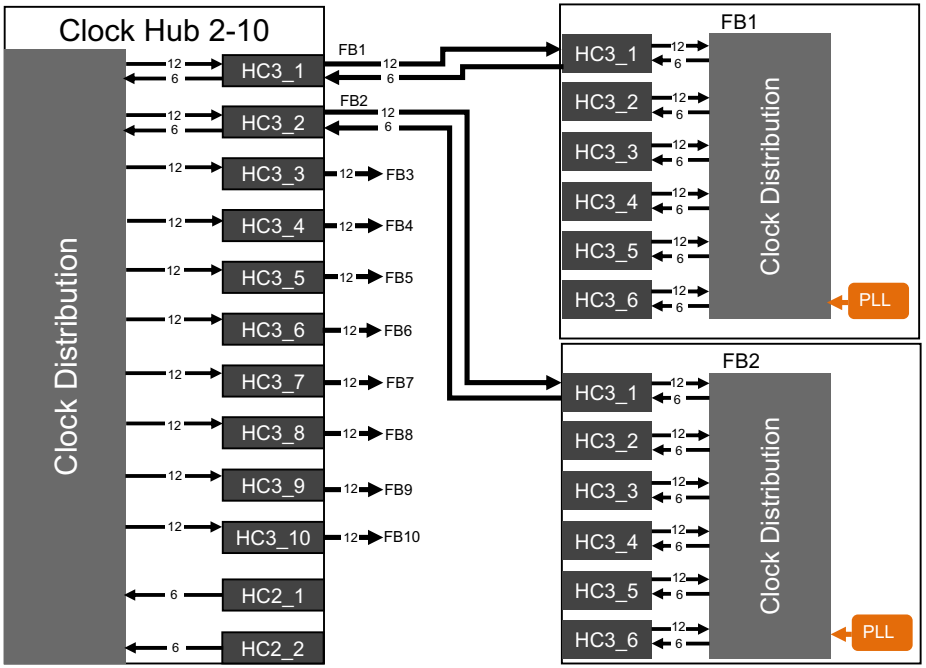
Clock Hub 2-10 is used to distribute clocks to more than 6 modules. The hub supports up to 10 clock destinations. It has 10 HC3 connectors of which the first two connectors, HC3_1 and HC3_2, are IN/OUT connectors, which can be used to drive clocks from HAPS-200 modules. Each connector (HC3_1 and HC3_2) can drive 6 clocks, in total 12 clocks to a hub. All these 12 clocks can be generated from a single HAPS-200 module or from two modules with 6 clocks each.

The remaining 8 HC3 connectors, HC3_3 to HC3_10, are output only and can be used to send clocks to other HAPS-200 or HAPS-100 modules using the HC3 to HC2 Adapter.

The hub has 2 HC2 inputs, HC2_1 and HC2_2, to drive 12 clocks from a HAPS-100 module.

All 12 clocks must be either generated by HAPS-200 or HAPS-100. The combination of 6 clocks from HAPS-200 and 6 clocks from HAPS-100 is not supported.

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Use case: 12 clocks shared across 10 modules.

In this scenario, the 6 clocks are generated from the FB1 and another 6 clocks are generated FB2. These clocks are shared among 10 HAPS-200 modules as shown in the above diagram.

Refer to Example section for TSS examples on using Clock Hub 2-10.

Modules vs Number of Clocks Table with Clock Hub 2-10

Number of Modules	Maximum Number of Clocks
Up to 10 Modules	12 clocks - If the clocks are generated from 2 modules. 6 clocks - If the clocks are generated from 1 module.
Up to 9 Modules	12 clocks - If the clocks are generated from 1 module.

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CHAPTER 6

Getting Started

Remember that the HAPS-200 12F module, as all electronic equipment, is sensitive to static discharge. Make sure that you're properly grounded whenever handling the system directly. Make sure to follow all safety precautions, see Important Safety Precautions, on page 12.

Special Instructions for Adding Extension Boards and Cables

Extension Boards and cables must be attached carefully! If connected improperly socket pins can be deformed or holding rail ends can become damaged.

Always consult with the extension board reference manual. Most extension boards are powered through the HapsTrak connector, but some may need external power.

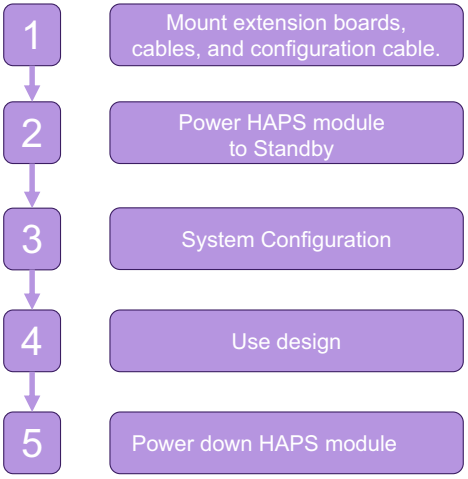
Important! Do not hot plug extension boards and cables. Always power down the HAPS-200 12F module when adding or removing extension boards and cables. There is a risk of damaging both the system and extension board if this instruction is not followed.

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Work Flow

Desktop mode

Follow these basic steps to get your design up and running on the HAPS-200 12F system.



1. Mount extension boards, cables, and configuration cable.

With the HAPS system powered off, connect the required extension boards, cables, and USB3 or QSFP28 cable according to your design.

2. Power on HAPS module

When power is applied to the module there is a boot sequence which may take a few seconds. During this phase the Power LED will blink green.

3. System configuration

Configure the system (for example, setting clocks and voltages) and FPGA design using hRun. See Configuration and Communication, on page 54.

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4. Use design

See the HAPS ProtoCompiler documentation.

5. Power down HAPS module

Press Power button for 5 seconds to power down.

Do not AC power cycle.

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CHAPTER 7

Configuration and Communication

HAPS-200 12F can be configured by either a USB connection or by one of the six QSFP-DD host interface connectors.

UMRBus®

USB3

The USB3 connector provides backward compatibility to early HAPS systems- Connect a USB cable to the host PC and the HAPS-200 12F module.

QSFP-DD

On a HAPS-200 12F module there are 6 QSFP-DD connectors (each with 2 HOSTs) for configuration and data streaming. The Universal HIC 200 card, with the help of QSFP28 cables, connects your host PC to up to 4 HAPS-200 modules.

Synchronous Reset

Global reset does not exist in HAPS-200 modules, only local reset lines (active low) for individual user FPGAs that are part of the System IP and controlled externally from *hRun*.

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LED Behavior

LED	Color	Behavior
USB3 HOST	Off	No USB cable is connected
	Red	USB connection error
	Amber	No driver loaded. USB cable connected.
	Green	High speed or low speed USB connection (USB 2.0 or less
	Blue	SuperSpeed USB connection (USB 3.0)
QSFP-DD HOST	Off	No QSFP cable is connected
	Green	Host is connected. Link to host is active.
FPGAs	Off	Off
	Red	Error
	Amber	Unconfigured
	Green	Configured

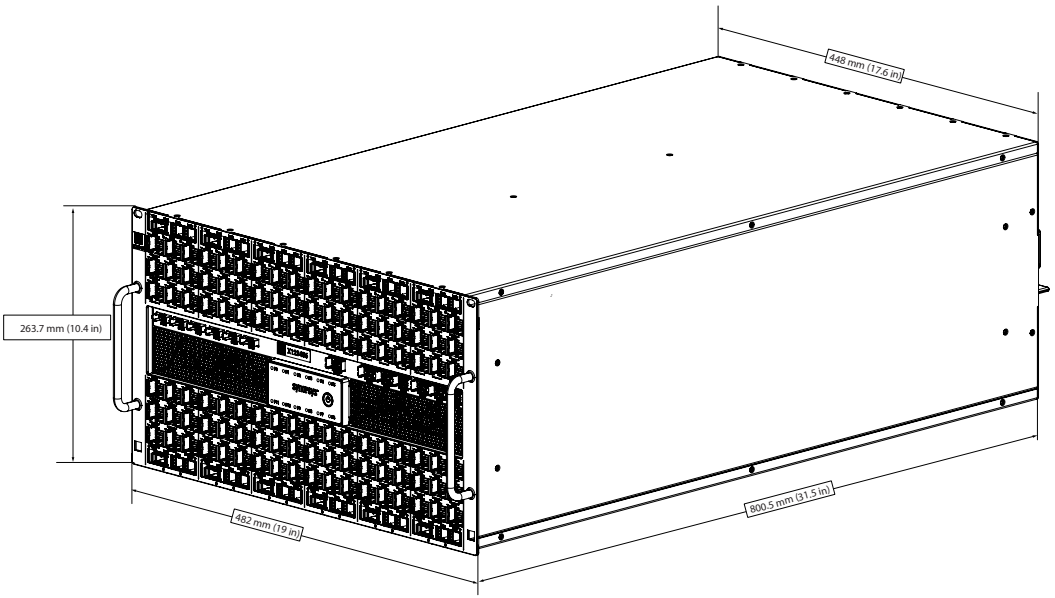
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CHAPTER 8

System

Dimensions

(LxWxH) 800.5 mm x 448 mm x 263.7 mm (31.5 in x17.6 in x 10.4 in)



Weight

- 75 kg (165.4 lb) (this weight does not include cables or accessories).
- 100 kg (220 lb) packaged gross weight.

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CHAPTER 9

ProtoCompiler TSS and PCF Examples

The Target System Specification (TSS) file is a Tcl file that defines the hardware configuration and the Partition Constraints File (PCF) controls the partitioning operation in HAPS ProtoCompiler. This section describes specific TSS and PCF commands for HAPS-200 12F.

For more information about TSS and PCF go to the *HAPS ProtoCompiler Documentation* on SolvNetPlus or in your ProtoCompiler installation.

PLL Source Clock Example

- Single HAPS-200 12F system
- 2 PLL clocks connected to all 12 FPGAs in HAPS-200 12F system
- Generic PLL command format

TSS

```
board_system_create -haps -name DefaultSystem

board_system_create -add HAPS200_12F_2 -name FB1

#Specifying the top io connection
board_system_configure -top_io { FB1.F4_X5}

# PLL specified clock source connected across all FPGAs.
board_system_configure -clk_src {PLL} -name myclk1 -frequency 1000

board_system_configure -clock {FB1.F0.CLK1} myclk1
```


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```
board_system_configure -clock {FB1.F1.CLK1} myclk1
board_system_configure -clock {FB1.F2.CLK1} myclk1
board_system_configure -clock {FB1.F3.CLK1} myclk1
board_system_configure -clock {FB1.F4.CLK1} myclk1
board_system_configure -clock {FB1.F5.CLK1} myclk1
board_system_configure -clock {FB1.F6.CLK1} myclk1
board_system_configure -clock {FB1.F7.CLK1} myclk1
board_system_configure -clock {FB1.F8.CLK1} myclk1
board_system_configure -clock {FB1.F9.CLK1} myclk1
board_system_configure -clock {FB1.F10.CLK1} myclk1
board_system_configure -clock {FB1.F11.CLK1} myclk1

board_system_configure -clk_src {PLL} -name myclk2 -frequency 2000
board_system_configure -clock {FB1.F0.CLK2} myclk2
board_system_configure -clock {FB1.F1.CLK2} myclk2
board_system_configure -clock {FB1.F2.CLK2} myclk2
board_system_configure -clock {FB1.F3.CLK2} myclk2
board_system_configure -clock {FB1.F4.CLK2} myclk2
board_system_configure -clock {FB1.F5.CLK2} myclk2
board_system_configure -clock {FB1.F6.CLK2} myclk2
board_system_configure -clock {FB1.F7.CLK2} myclk2
board_system_configure -clock {FB1.F8.CLK2} myclk2
board_system_configure -clock {FB1.F9.CLK2} myclk2
board_system_configure -clock {FB1.F10.CLK2} myclk2
board_system_configure -clock {FB1.F11.CLK2} myclk2
```


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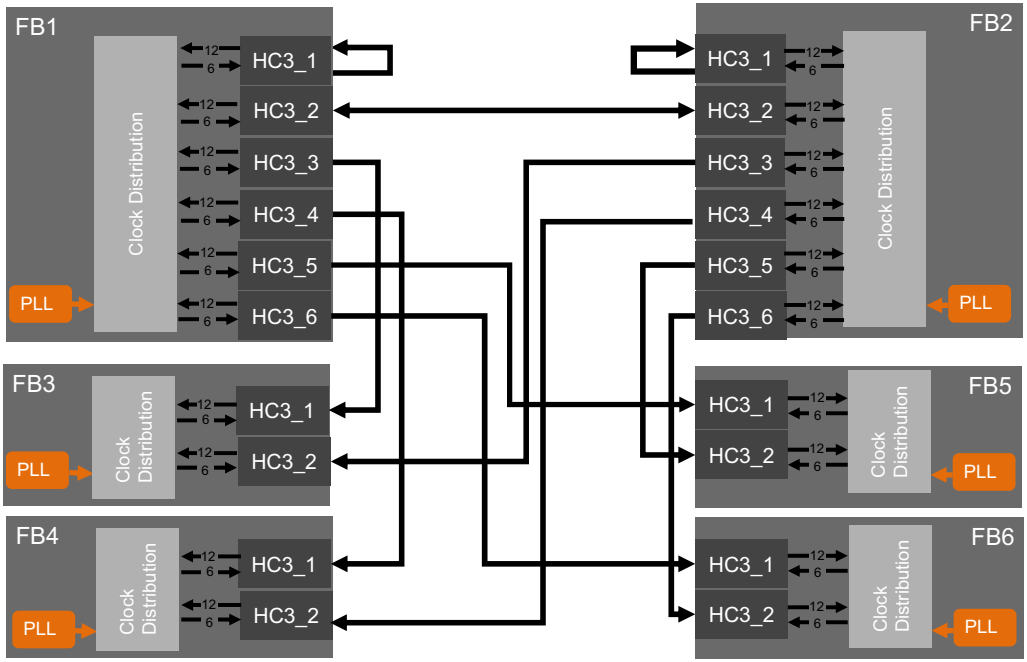
PCF

```
assign_global_net designclk1 myclk1

assign_global_net designclk2 myclk2
```

Multi-system (6 systems) HAPS-200 12F Board PLL Clock Example

- 6 HAPS-200 12F systems
- System 1 (FB1) generates 6 clocks and system 2 (FB2) generates 6 clocks.
- The 12 clocks are connected to all 6 HAPS-200 12F systems



TSS

```
board_system_create -haps -name DefaultSystem

board_system_create -add HAPS200_12F_2 -name FB1

board_system_create -add HAPS200_12F_2 -name FB2
```


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```
board_system_create -add HAPS200_12F_2 -name FB3

board_system_create -add HAPS200_12F_2 -name FB4

board_system_create -add HAPS200_12F_2 -name FB5

board_system_create -add HAPS200_12F_2 -name FB6

#FB1 generated clocks

board_system_configure -clk_src FB1.PLL -name myclk1
-frequency 1234

board_system_configure -clk_src FB1.PLL -name myclk2
-frequency 1234

board_system_configure -clk_src FB1.PLL -name myclk3
-frequency 1234

board_system_configure -clk_src FB1.PLL -name myclk4
-frequency 1234

board_system_configure -clk_src FB1.PLL -name myclk5
-frequency 1234

board_system_configure -clk_src FB1.PLL -name myclk6
-frequency 1234


#FB2 generated clocks

board_system_configure -clk_src FB2.PLL -name myclk7
-frequency 1234

board_system_configure -clk_src FB2.PLL -name myclk8
-frequency 1234

board_system_configure -clk_src FB2.PLL -name myclk9
-frequency 1234

board_system_configure -clk_src FB2.PLL -name myclk10
-frequency 1234

board_system_configure -clk_src FB2.PLL -name myclk11
-frequency 1234

board_system_configure -clk_src FB2.PLL -name myclk12
-frequency 1234
```


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```
#Specify the HC3 connectors
#loopback connections
board_system_configure -connect {FB1.HC3_1 FB1.HC3_1}
board_system_configure -connect {FB2.HC3_1 FB2.HC3_1}

#Other connectors
board_system_configure -connect {FB1.HC3_2 FB2.HC3_2}

board_system_configure -connect {FB1.HC3_3 FB3.HC3_1}
board_system_configure -connect {FB2.HC3_3 FB3.HC3_2}

board_system_configure -connect {FB1.HC3_4 FB4.HC3_1}
board_system_configure -connect {FB2.HC3_4 FB4.HC3_2}

board_system_configure -connect {FB1.HC3_5 FB5.HC3_1}
board_system_configure -connect {FB2.HC3_5 FB5.HC3_2}

board_system_configure -connect {FB1.HC3_6 FB6.HC3_1}
board_system_configure -connect {FB2.HC3_6 FB6.HC3_2}

#Connecting to FPGA Global clocks
for {set a 1} {$a < 13} {incr a} {
if {$a < 7} {
board_system_configure -clock FB1.F0.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a
board_system_configure -clock FB1.F1.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a
```


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```
board_system_configure -clock FB1.F2.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a

board_system_configure -clock FB1.F3.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a

board_system_configure -clock FB1.F4.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a

board_system_configure -clock FB1.F5.CLK$a myclk$a
-through /FB1/HC3_1/IN/CLK$a

} else {

board_system_configure -clock FB1.F0.CLK$a myclk$a

board_system_configure -clock FB1.F1.CLK$a myclk$a

board_system_configure -clock FB1.F2.CLK$a myclk$a

board_system_configure -clock FB1.F3.CLK$a myclk$a

board_system_configure -clock FB1.F4.CLK$a myclk$a

board_system_configure -clock FB1.F5.CLK$a myclk$a

}

If {$a < 7} {

board_system_configure -clock FB2.F0.CLK$a myclk$a

board_system_configure -clock FB2.F1.CLK$a myclk$a

board_system_configure -clock FB2.F2.CLK$a myclk$a

board_system_configure -clock FB2.F3.CLK$a myclk$a

board_system_configure -clock FB2.F4.CLK$a myclk$a

board_system_configure -clock FB2.F5.CLK$a myclk$a

} else {

Set pin [expr {$a-6}]

board_system_configure -clock FB2.F0.CLK$a myclk$a

-through /FB2/HC3_1/IN/CLK$pin

board_system_configure -clock FB2.F1.CLK$a myclk$a

-through /FB2/HC3_1/IN/CLK$pin
```


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```
board_system_configure -clock FB2.F2.CLK$a myclk$a
-through /FB2/HC3_1/IN/CLK$pin
board_system_configure -clock FB2.F3.CLK$a myclk$a
-through /FB2/HC3_1/IN/CLK$pin
board_system_configure -clock FB2.F4.CLK$a myclk$a
-through /FB2/HC3_1/IN/CLK$pin
board_system_configure -clock FB2.F5.CLK$a myclk$a
-through /FB2/HC3_1/IN/CLK$pin
}

board_system_configure -clock FB3.F0.CLK$a myclk$a
board_system_configure -clock FB3.F1.CLK$a myclk$a
board_system_configure -clock FB3.F2.CLK$a myclk$a
board_system_configure -clock FB3.F3.CLK$a myclk$a
board_system_configure -clock FB3.F4.CLK$a myclk$a
board_system_configure -clock FB3.F5.CLK$a myclk$a

board_system_configure -clock FB4.F0.CLK$a myclk$a
board_system_configure -clock FB4.F1.CLK$a myclk$a
board_system_configure -clock FB4.F2.CLK$a myclk$a
board_system_configure -clock FB4.F3.CLK$a myclk$a
board_system_configure -clock FB4.F4.CLK$a myclk$a
board_system_configure -clock FB4.F5.CLK$a myclk$a

board_system_configure -clock FB5.F0.CLK$a myclk$a
board_system_configure -clock FB5.F1.CLK$a myclk$a
board_system_configure -clock FB5.F2.CLK$a myclk$a
```


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```
board_system_configure -clock FB5.F3.CLK$a myclk$a
board_system_configure -clock FB5.F4.CLK$a myclk$a
board_system_configure -clock FB5.F5.CLK$a myclk$a

board_system_configure -clock FB6.F0.CLK$a myclk$a
board_system_configure -clock FB6.F1.CLK$a myclk$a
board_system_configure -clock FB6.F2.CLK$a myclk$a
board_system_configure -clock FB6.F3.CLK$a myclk$a
board_system_configure -clock FB6.F4.CLK$a myclk$a
board_system_configure -clock FB6.F5.CLK$a myclk$a
}
```

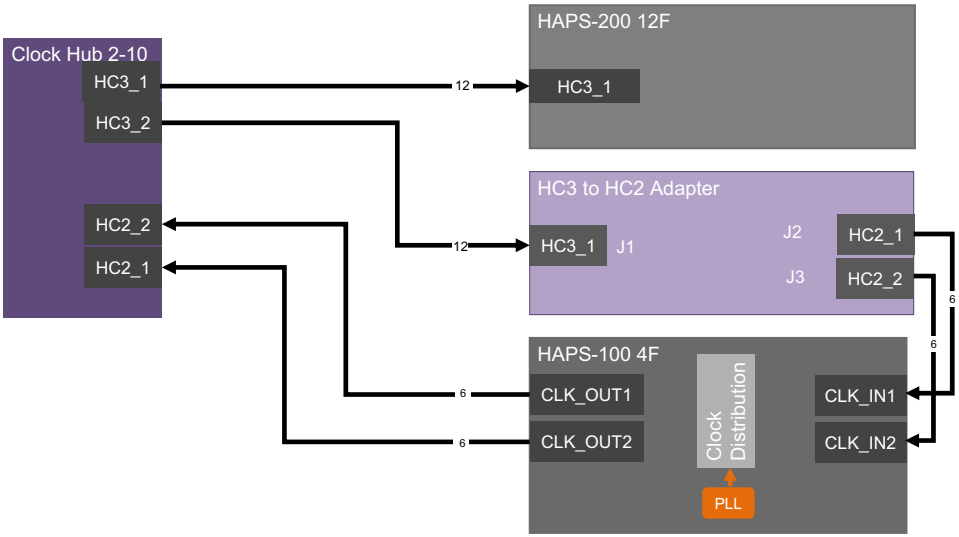
PCF

The PCF commands should be in the same format as in prior examples.

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Mixed System Clocks Example

In this example,12 clocks are generated from a HAPS-100 system and distributed to a HAPS-200 12F system and then back to the original HAPS-100 system.



TSS

```
board_system_create -haps -name DefaultSystem

board_system_create -add HAPS200_12F_2 -name FB1

board_system_create -add HAPS100_4F -name MB1


#Declare PLL clocks from MB1

board_system_configure -clk_src MB1.PLL -name myclk1
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk2
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk3
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk4
-frequency 1234
```


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```
board_system_configure -clk_src MB1.PLL -name myclk5
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk6
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk7
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk8
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk9
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk10
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk11
-frequency 1234

board_system_configure -clk_src MB1.PLL -name myclk12
-frequency 1234


#Instantiate clock hub and clocks HAPS-100 to HAPS-200 using HC2

connectorboard_system_create -interconnect -manual CLK_HUB2X10_HC3
-name clk_hub
-connector {.HC2_1(MB1.CLK_OUT1).HC2_2(MB1.CLK_OUT2)}

#Connect HC3 cable from HUB to HAPS-200 system

board_system_configure -connect {clk_hub.HC3_1 FB1.HC3_1}

#Instantiate HC2X2_ADPTR_HC3 and HC3 connector

board_system_create -interconnect -manual HC2X2_ADPTR_HC3
-name clk_hc2 -connector {.HC3_1(clk_hub.HC3_2)}


#Connect HC2 cables from adapter to HAPS-100

board_system_configure -connect {clk_hc2.HC2_1 MB1.CLK_IN1}

board_system_configure -connect {clk_hc2.HC2_2 MB1.CLK_IN2}
```


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```
#Connect FPGA Global clocks

for {set a 1} {$a < 13} {incr a} {

if {$a < 7} {

board_system_configure -clock FB1.F0.CLK$a myclk$a

board_system_configure -clock FB1.F1.CLK$a myclk$a

board_system_configure -clock FB1.F2.CLK$a myclk$a

board_system_configure -clock FB1.F3.CLK$a myclk$a

board_system_configure -clock FB1.F4.CLK$a myclk$a

board_system_configure -clock FB1.F5.CLK$a myclk$a


board_system_configure -clock MB1.uA.CLK${a} myclk${a}
-through /MB1/CLK_IN1/CLK$a

board_system_configure -clock MB1.uB.CLK${a} myclk${a}
-through /MB1/CLK_IN1/CLK$a

board_system_configure -clock MB1.uC.CLK${a} myclk${a}
-through /MB1/CLK_IN1/CLK$a

board_system_configure -clock MB1.uD.CLK${a} myclk${a}
-through /MB1/CLK_IN1/CLK$a

} else {

set b [expr {$a-6}]

set k [expr {$a+6}]

board_system_configure -clock FB1.F0.CLK$k myclk$a

board_system_configure -clock FB1.F1.CLK$k myclk$a

board_system_configure -clock FB1.F2.CLK$k myclk$a

board_system_configure -clock FB1.F3.CLK$k myclk$a

board_system_configure -clock FB1.F4.CLK$k myclk$a

board_system_configure -clock FB1.F5.CLK$k myclk$a

board_system_configure -clock MB1.uA.CLK${a} myclk${a}
-through /MB1/CLK_IN2/CLK$b
```


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```
board_system_configure -clock MB1.uB.CLK${a} myclk${a}
-through /MB1/CLK_IN2/CLK$b

board_system_configure -clock MB1.uC.CLK${a} myclk${a}
-through /MB1/CLK_IN2/CLK$b

board_system_configure -clock MB1.uD.CLK${a} myclk${a}
-through /MB1/CLK_IN2/CLK$b

}

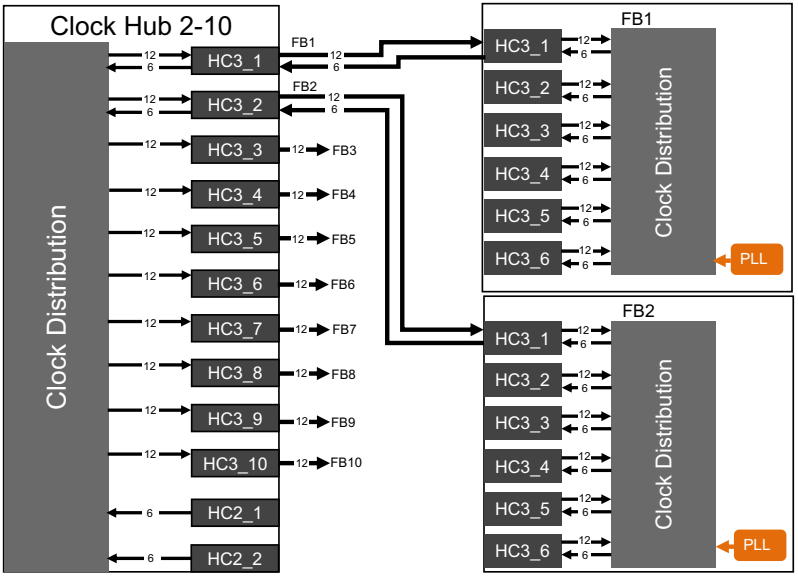
}
```

PCF

The PCF commands are the same format as in prior examples.

Clock Hub Example

- 12 clocks shared across 10 HAPS-200 12F systems.
- 6 PLL clocks are generated from FB1
- 6 PLL clocks are generated from FB2
- Clocks are shared among 10 HAPS-200 12F systems



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TSS

```
board_system_create -haps -name DefaultSystem

board_system_create -add HAPS200_12F_2 -name FB1

#Similar syntax for remaining 9 systems

board_system_create -add HAPS200_12F_2 -name FB10

#Similar syntax for remaining 9 systems

#Declare PLL clocks from FB1

board_system_configure -clk_src FB1.PLL1.CLK1 -name myclk_s1_1
-frequency 1000

board_system_configure -clk_src FB2.PLL1.CLK1 -name myclk_s2_1
-frequency 1000

board_system_configure -clk_src FB1.PLL2.CLK1 -name myclk_s1_4
-frequency 1000

board_system_configure -clk_src FB2.PLL2.CLK1 -name myclk_s2_4
-frequency 1000

board_system_configure -clk_src FB1.PLL1.CLK2 -name myclk_s1_2
-frequency 1000

board_system_configure -clk_src FB2.PLL1.CLK2 -name myclk_s2_2
-frequency 1000

board_system_configure -clk_src FB1.PLL2.CLK2 -name myclk_s1_5
-frequency 1000

board_system_configure -clk_src FB2.PLL2.CLK2 -name myclk_s2_5
-frequency 1000

board_system_configure -clk_src FB1.PLL1.CLK3 -name myclk_s1_3
-frequency 1000

board_system_configure -clk_src FB2.PLL1.CLK3 -name myclk_s2_3
-frequency 1000

board_system_configure -clk_src FB1.PLL2.CLK3 -name myclk_s1_6
-frequency 1000

board_system_configure -clk_src FB2.PLL2.CLK3 -name myclk_s2_6
-frequency 1000

#Declare the HUB and driving clocks from FB1 and FB2 using HC3
```


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```
board_system_create -interconnect -manual CLK_HUB2X10_HC3
-name clk_hub -connector {.HC3_1(FB1.HC3_1) .HC3_2(FB2.HC3_1)}

#Connect HC3 cable from HUB to remaining HAPS-200 systems

board_system_configure -connect {clk_hub.HC3_3 FB3.HC3_1}

board_system_configure -connect {clk_hub.HC3_4 FB4.HC3_1}

board_system_configure -connect {clk_hub.HC3_5 FB5.HC3_1}

board_system_configure -connect {clk_hub.HC3_6 FB6.HC3_1}

board_system_configure -connect {clk_hub.HC3_7 FB7.HC3_1}

board_system_configure -connect {clk_hub.HC3_8 FB8.HC3_1}

board_system_configure -connect {clk_hub.HC3_9 FB9.HC3_1}

board_system_configure -connect {clk_hub.HC3_10 FB10.HC3_1}

#Connect to FPGA Global clocks set clk_s 6

set h200_sys_s 10

for {set b 1} {$b < ($h200_sys_s+1)} {incr b} {
for {set a 1} {$a < ($clk_s+1)} {incr a} {
for {set c 0} {$c < 12} {incr c} {


if {$b == 1} {

board_system_configure -clock FB$b.F${c}.CLK${a} myclk_s1_${a}
-through /FB1/HC3_1/IN/CLK$a

   } else {

board_system_configure -clock FB$b.F${c}.CLK${a} myclk_s1_${a}

   }

}

}

}

for {set b 1} {$b < ($h200_sys_s+1)} {incr b} {
for {set a 7} {$a < (2*$clk_s+1)} {incr a} {
for {set c 0} {$c < 12} {incr c} {
```


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```
   set j [expr {$a-6}]

   set bias [expr {$a+6}] if {$b == 2} {

   board_system_configure -clock FB$b.F${c}.CLK$bias myclk_s2_$j
-through /FB2/HC3_1/IN/CLK$a


   } else {

   board_system_configure -clock FB$b.F${c}.CLK$bias myclk_s2_$j

   }

}

}

}
```

FPGA CLOCKS Examples

Single System

2 FPGA clocks are generated from F0 FPGA of the HAPS-200 12F system. The 2 FPGA clocks are shared to all the FPGAs on the single system.

TSS

```
#Declare FPGA clocks

board_system_configure -clk_src FB1.F0.CLK_SRC1 -frequency 12345
-name fpgaclk1

board_system_configure -clk_src FB1.F0.CLK_SRC5 -frequency 12345
-name fpgaclk2

#Connecting to valid GCLKS of all the FPGAs

board_system_configure -clock FB1.F0.CLK1 fpgaclk1
board_system_configure -clock FB1.F0.CLK2 fpgaclk2

board_system_configure -clock FB1.F1.CLK1 fpgaclk1
board_system_configure -clock FB1.F1.CLK2 fpgaclk2

board_system_configure -clock FB1.F2.CLK1 fpgaclk1
board_system_configure -clock FB1.F2.CLK2 fpgaclk2
```


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```
board_system_configure -clock FB1.F3.CLK1 fpgaclk1
board_system_configure -clock FB1.F3.CLK2 fpgaclk2

board_system_configure -clock FB1.F4.CLK1 fpgaclk1
board_system_configure -clock FB1.F4.CLK2 fpgaclk2

board_system_configure -clock FB1.F5.CLK1 fpgaclk1
board_system_configure -clock FB1.F5.CLK2 fpgaclk2

board_system_configure -clock FB1.F6.CLK1 fpgaclk1
board_system_configure -clock FB1.F6.CLK2 fpgaclk2

board_system_configure -clock FB1.F7.CLK1 fpgaclk1
board_system_configure -clock FB1.F7.CLK2 fpgaclk2

board_system_configure -clock FB1.F8.CLK1 fpgaclk1
board_system_configure -clock FB1.F8.CLK2 fpgaclk2

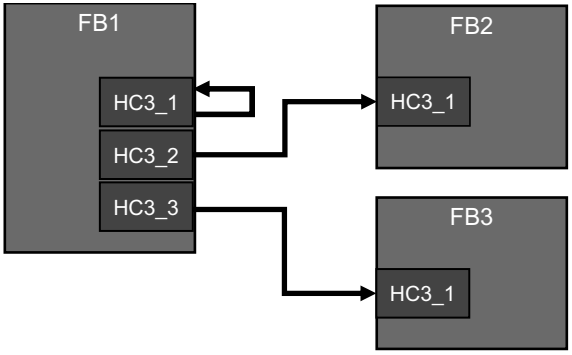
board_system_configure -clock FB1.F9.CLK1 fpgaclk1
board_system_configure -clock FB1.F9.CLK2 fpgaclk2

board_system_configure -clock FB1.F10.CLK1 fpgaclk1
board_system_configure -clock FB1.F10.CLK2 fpgaclk2

board_system_configure -clock FB1.F11.CLK1 fpgaclk1
board_system_configure -clock FB1.F11.CLK2 fpgaclk2
```

Multisystem without a Clock Hub

2 FPGA clocks generated from a HAPS-200 system and shared to 2 HAPS-200 12F systems.



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TSS

```
#FPGA clock declaration

board_system_configure -clk_src FB1.F0.CLK_SRC1 -frequency 12345
-name fpgaclk1

board_system_configure -clk_src FB1.F0.CLK_SRC5 -frequency 12345
-name fpgaclk2

#Clock connections from first system to others and loopback

board_system_configure -connect {FB1.HC3_1 FB1.HC3_1}
board_system_configure -connect {FB1.HC3_2 FB2.HC3_1}
board_system_configure -connect {FB1.HC3_3 FB3.HC3_1}

#Connect to the valid GCLKs of the first system
board_system_configure -clock FB1.F0.CLK1 fpgaclk1
-through FB1/HC3_1/IN/CLK1

board_system_configure -clock FB1.F0.CLK2 fpgaclk2 -through
FB1/HC3_1/IN/CLK2

#Similar commands for the remaining FPGAs in the FB1 system

#Connect to the valid GCLKs of the second system
board_system_configure -clock FB2.F0.CLK1 fpgaclk1
board_system_configure -clock FB2.F0.CLK2 fpgaclk2

#Similar commands for the remaining FPGAs in the FB2 system

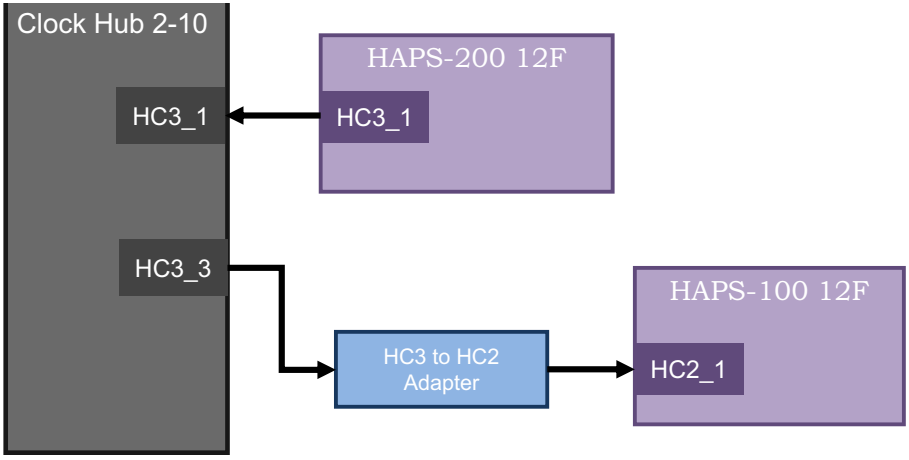
#Connect to the valid GCLKs of the third system
board_system_configure -clock FB3.F0.CLK1 fpgaclk1
board_system_configure -clock FB3.F0.CLK2 fpgaclk2

#Similar commands for the remaining FPGAs in the FB3 system
```


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Mixed System Clock Example HAPS-200 12F & HAPS-100 12F

In this example, HAPS-200 12F is connected to a Clock Hub 2-10. An HC3 to HC2 Adapter is used to connect the clock hub to HAPS-200 12F.




```
board_system_create -add HAPS200_12F_2 -name FB1
board_system_create -add HAPS100_12F -name FB2 -speed_grade {-2-e}

## HAPS-200 12F provides clocks to HAPS-100 12F

board_system_create -interconnect -manual CLK_HUB2X10_HC3
-name clk_hub -connector {.HC3_1(FB1.HC3_1)}

board_system_create -interconnect -manual HC2X2_ADAPTER_HC3
-name clkhc3hc2_1 -connector {.HC3_1(clk_hub.HC3_3)}

board_system_configure -connect {clkhc3hc2_1.HC2_1 FB2.CLK_IN1}

board_system_configure -clk_src FB1.PLL1.CLK1 -frequency 1230
-name dutclk1

board_system_configure -clock {FB1.F0.CLK1} dutclk1
-through /FB1/HC3_1/IN/CLK1

board_system_configure -clock {FB2.uF1.CLK1} dutclk1
```


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